Practical No 05. Adapter Based Views

A) Develop an Android application using ListView with ArrayAdapter and implement item click handling.

Project Name: ListViewDemo

Program Code:

Filename: activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="10dp">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Programming Languages"
        android:textSize="24sp"
        android:textStyle="bold"
        android:padding="10dp" />

    <ListView
        android:id="@+id/listView"
        android:layout_width="match_parent"
        android:layout_height="match_parent" />

</LinearLayout>
```

Filename: :- Main Activity.java

```
package com.example.listviewdemo;

import android.os.Bundle;
import android.view.View;
import android.widget.AdapterView;
import android.widget.ArrayAdapter;
import android.widget.ListView;
```

```
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    private ListView listView;
    private final String[] languages = {
            "Java",
            "Python",
            "C",
            "C++",
            "Kotlin",
            "PHP",
            "JavaScript",
            "Swift"
    };

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        listView = findViewById(R.id.listView);

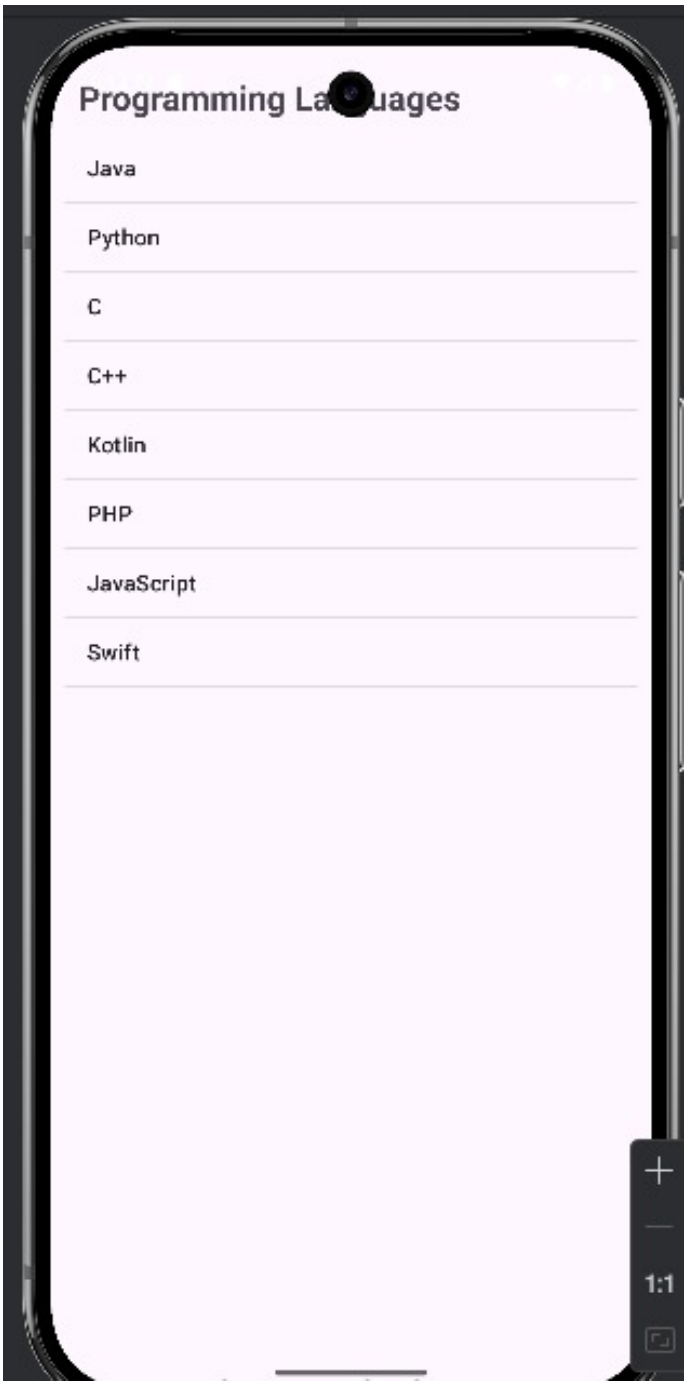

        ArrayAdapter<String> adapter = new ArrayAdapter<>(
                this,
                android.R.layout.simple_list_item_1,
                languages
        );

        listView.setAdapter(adapter);

        listView.setOnItemClickListener(new AdapterView.OnItemClickListener() {
            @Override
            public void onItemClick(AdapterView<?> parent, View view, int position, long id) {
                String item = languages[position];
                Toast.makeText(
                        MainActivity.this,
                        "Selected: " + item,
                        Toast.LENGTH_SHORT
                ).show();
            }
        });
```

```
    }
}
```

Output: ListViewDemo



B) Develop an Android application using GridView with Custom Adapter and image selection handling.